EART11200: Natural Scientist's Toolkit



Some basic tools to get you started

Scientific Notation

A simple way to write very large or very small numbers:

$10{,}000{,}000{,}000{,}000 = 10^{13} = 1E13$

$0.00000000001 = 10^{-12} = 1E\text{-}12$

If you are multiplying 2 powers of 10, add the powers (superscript numbers, # after E)

$(3 \times 10^{13}) \times (2 \times 10^{-12}) = 6 \times 10^{1} = 60$

If dividing, subtract the powers:

$1E13/1E\text{-}12 = 1E25$ [13-(-12) = 25]

NB $10^{0} = 1$

Law of indices

You have just seen an example of the law of indices (index or power: b^3)

Indices can be added or subtracted (when numbers are multiplied or divided), so long as the base number (b above) is the same – for scientific notation/prefixes, b is 10.

Indices cannot be combined in this way if the base number changes:

a^2b^4 cannot be simplified, but

$a^2b^4 \times a^5b^6c = a^7b^{10}c$

Squares, cubes, roots and indices

Provided the base number is the same there are some further tricks with indices.

$(a^3)^3 = a^3 \times a^3 \times a^3 = a^9$ (multiply)

The square root of a number can be written $a^{1/2}$ while $a^{1/3}$ is a cube root. These fractional indices are used in just the same way as whole numbers.

Read supporting material (all_maths) if you are unfamiliar with indices, or ask in the practical class.

Some example prefixes

A.1 Prefixes

Here are some prefixes that are often used when writing scientific numbers.

10	deka (da)	10^{-1}	deci (d)
10^2	hecta (h)	10^{-2}	centi (c)
10^3	kilo (k)	10^{-3}	milli (m)
10^6	mega (M)	10^{-6}	micro (μ)
10^9	giga (G)	10^{-9}	nano (n)
10^{12}	tera (T)	10^{-12}	pico (p)
10^{15}	peta (P)	10^{-15}	femto (f)
10^{18}	exa (E)	10^{-18}	atto (a)

E-7 / E-9 = E2 = x10^2 = 100
(-7) – (-9) = 2

E24 x 1000 = E27
E27 / E18 = E9

- Wavelength of green light: 5 E-7 m. How many nanometres? 500
- Solar constant: ~1300 W m^{-2}. How many kW m^{-2}? 1.3
- Size of a cloud drop 1E-5 m. How many micro-metres? 10
- Energy consumption in the UK: 6E11 W. How many TW? 0.6
- Mass of earth: 6E24 kg. How many exa grams? (tricky) 6E9
- Mass of hydrogen atom: 1.66E-27 kg. How many atto grams? (tricky) 1.66E-6